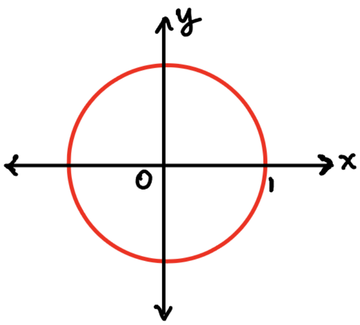
Ms. Welles **Alg 2 Trig/Honors**

<u>Chp 6.1 The Unit Circle</u>

I. The Unit Circle

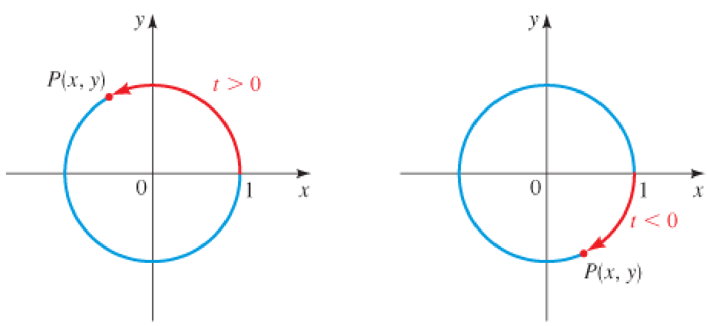
The unit circle is the circle of __________ centered at the ___________in the xy-plane . Its equation is _________________.



II. Terminal Points on the Unit Circle

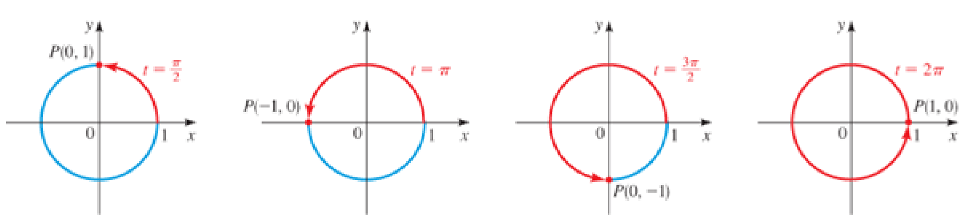
Suppose t is a real number. If $t \geq 0$, let's mark off a distance t along the unit circle, starting at __________ and moving in a ________________________ direction if *t* is positive, or in a ________________ direction if *t* is negative. In this way we arrive at a point $P(x, y)$, called the ___________________________ determined by the real number t.

The circumference of the unit circle is $C =$______. To move a point halfway around the unit circle, it travels a distance of _______ . To move a quarter of the distance around the circle, it travels a distance of ________.



(a) Terminal point $P(x, y)$ determined by $t > 0$ (b) Terminal point $P(x, y)$ determined by $t < 0$

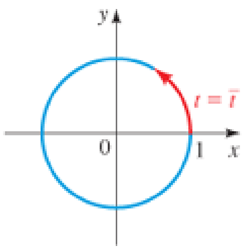
Terminal points determined by $t = \frac{\pi}{2},\ t = \pi,\ t = \frac{3\pi}{2},\ and\ t = 2\pi$.



III. The Reference Number

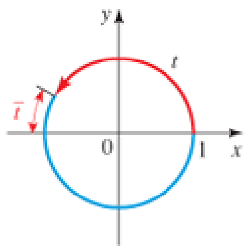
Let t be a real number. The **reference number** $\bar{t}$ associated with t is the **shortest distance** along the unit circle between ______________________ determined by t and the ___________ .To find the reference number $\bar{t}$ it is helpful to know the quadrant in which the terminal point determined by t lies . If the terminal point lies in quadrants **I or IV**, where x is positive, we find $\bar{t}$ by moving along the circle to the positive x-axis . If it lies in quadrants **II or III**, where x is negative, we find $\bar{t}$ by moving along the circle to the negative x-axis.

Quadrant I



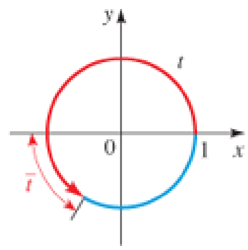
$\bar{t}$ = __________

Quadrant II



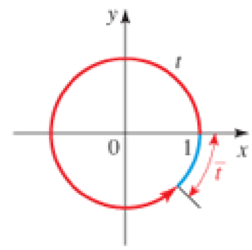
$\bar{t}$ = __________

Quadrant III



$\bar{t}$ = __________

Quadrant IV



$\bar{t}$ = _____________

<u>**Using Reference Numbers To Find Terminal Points**</u>

Steps for finding the terminal point *P* determined by any value of *t*

1. Find the **reference number** $\bar{t}$.
2. Find the **terminal point** Q(a, b) determined by $\bar{t}$.
3. The terminal point determined by ***t*** is P(±a, ±b) , where the **signs are chosen** according to the **quadrant** in which this terminal point lies.

Since the circumference of the unit circle is 2π, the terminal point determined by t is the same as that determined by $t + 2\pi$ or $t - 2\pi$. In general, we can add or subtract 2π any number of times without changing the terminal point determined by t.

Example 1: Show that the point is on the unit circle.

a. $\left(-\frac{5}{7},\ -\frac{2\sqrt{6}}{7}\right)$

b. $\left(\frac{3}{4},\ \frac{-\sqrt{7}}{4}\right)$

Example 2: Find the missing coordinate of P, using the fact that P lies on the unit circle in the given quadrant.

Coordinates	Quadrant	Coordinates	Quadrant
a. $P\left(-\frac{3}{5},\quad\right)$	III	b. $P\left(\quad, \frac{-2}{7}\right)$	IV

Example 3: The point P is on the unit circle. Find $P(x, y)$ from the given information.

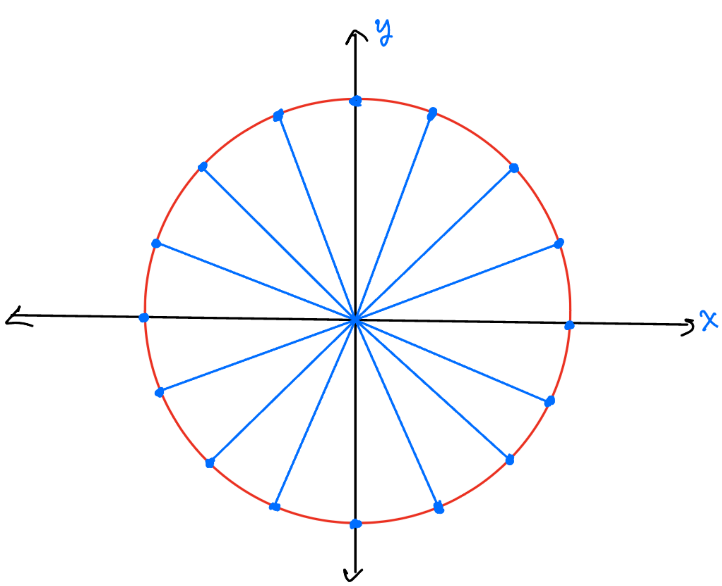
a. The y-coordinate of P is $-\frac{3}{5}$, and the x-coordinate is positive.

b. The x-coordinate of P is positive, and the y-coordinate of P is $\frac{-\sqrt{5}}{5}$

c. The x-coordinate of P is $-\frac{2}{5}$, and P lies above the x-axis.

SPECIAL TERMINAL POINTS

Example 4: Find the terminal points determined by the special values of t.



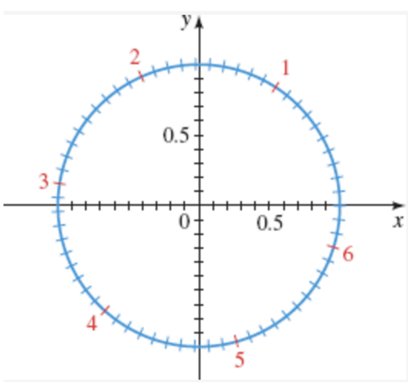
Exercise 5: Find the reference number for each value of *t*

$t = 9\pi$ b. $t = -\frac{41\pi}{4}$ c. $t = 6$ d. $t = -7$

Exercise 6: Find the terminal point $P(x, y)$ on the unit circle determined by the given value of *t*.

a. $t = 5\pi$ b. $t = -\frac{11\pi}{3}$ c. $t = \frac{5\pi}{2}$ d. $t = \frac{31\pi}{6}$

Example 7: The unit circle is graphed in the figure below. Use the figure to find the terminal point determined by the real number *t*, with coordinates rounded to one decimal place.



a. $t = 2.5$ b. $t = -1.1$ c. $t = 4.2$

Example 8: Suppose that the terminal point determined by *t* is the point $\left(\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find the terminal point determined by each of the following.

a. $-t$ b. $\pi - t$ c. $2\pi + t$